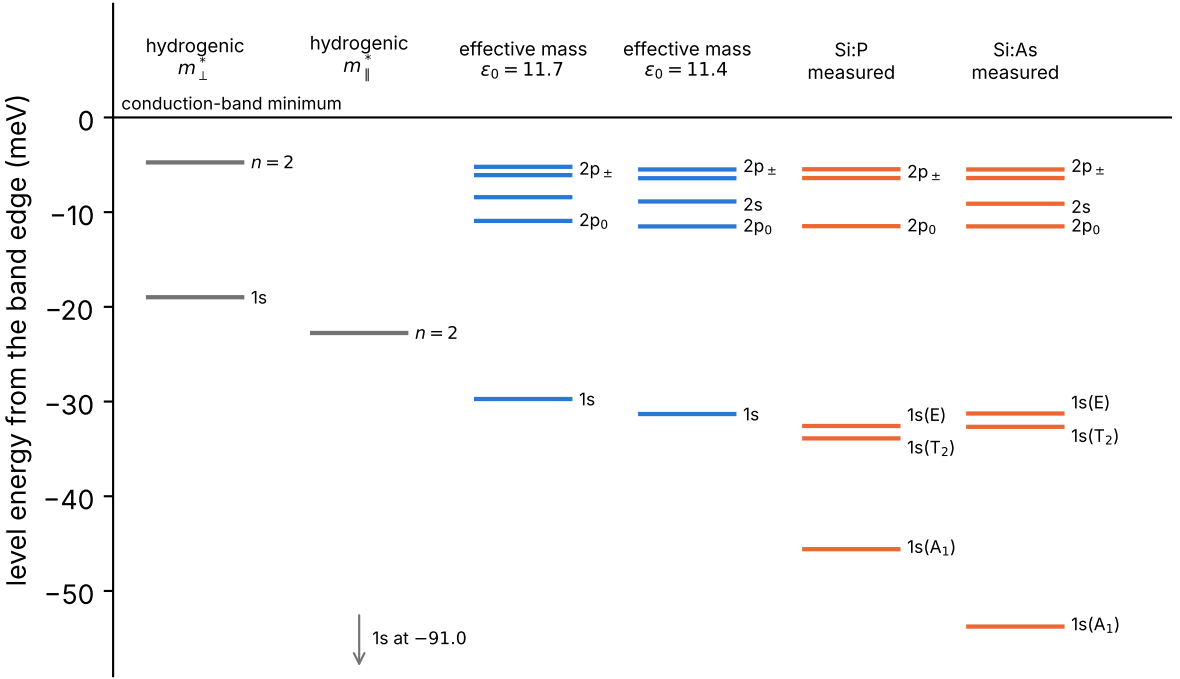
hydrogenic $m^*_\perp$
hydrogenic $m^*_\parallel$
effective mass $\epsilon_0 = 11.7$
effective mass $\epsilon_0 = 11.4$
Si:P measured
Si:As measured
conduction-band minimum
level energy from the band edge (meV)
0
−10
−20
−30
−40
−50
$n = 2$
1s
$n = 2$
1s at −91.0
$2p_\pm$
$2p_0$
1s
$2p_\pm$
2s
$2p_0$
1s
$2p_\pm$
$2p_0$
1s(E)
$1s(T_2)$
$1s(A_1)$
$2p_\pm$
2s
$2p_0$
1s(E)
$1s(T_2)$
$1s(A_1)$